

A Local Compact-Chart Criterion for C^* -Algebras with UBCP

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June 17, 2026

Source Problem

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving that a commutative unital C^* -algebra $C(K)$ has BCP/UBCP exactly when its pure state space, i.e. K , has a countable π -basis, the authors ask for a noncommutative version:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

Statement

The previous continuous-trace packet proves the positive theorem when the whole algebra is locally a continuous field of compact operators. The obstruction to extending that proof is that an element of a larger algebra may act on a continuous-trace ideal only as a multiplier. The following criterion isolates exactly the extra local regularity that is needed.

Definition 1. *Let A be a C^* -algebra. A compact-norming chart for A consists of:*

1. a nonempty open subset $O \subset \text{Prim}(A)$, with corresponding ideal I_O ;
2. a separable nonzero Hilbert space H_O ;
3. a $C_0(O)$ -isomorphism

$$\Phi_O : I_O \longrightarrow C_0(O, \mathcal{K}(H_O));$$

4. for every $a \in A$, a bounded norm-continuous function

$$\widehat{a}_O : O \longrightarrow \mathcal{K}(H_O)$$

such that $\widehat{b}_O = \Phi_O(b)$ for $b \in I_O$, and

$$\|a + P\| = \|\widehat{a}_O(P)\| \quad (P \in O).$$

Here $a + P$ denotes the image of a in the primitive quotient associated with P .

We say that A has a *countable compact-norming chart π -basis* if there are compact-norming charts $(O_n, \Phi_n, H_n)_{n \geq 1}$ such that (O_n) is a π -basis of $\text{Prim}(A)$.

Theorem 1. *Let A be a C^* -algebra. Assume that A has a countable compact-norming chart π -basis. Then A has UBCP. Consequently, for every C^* -algebra satisfying this chart hypothesis, the following are equivalent:*

1. A has BCP;
2. A has UBCP;
3. $\text{Prim}(A)$ has a countable π -basis.

Idea of the Proof

The chart hypothesis says that on countably many primitive-open patches, every ambient element of A is not merely a multiplier of a continuous-trace ideal: it is represented by a norm-continuous compact-operator section. That is exactly what the continuous-trace bump proof needs.

Given $a \in S_A$, the set where $\|a + P\| > 3/4$ is open. Choose a chart O_n inside this set. In that chart, $\widehat{a}_n$ is a continuous compact-operator field. At one point P_0 , choose a countable net direction $T \in \mathcal{K}(H_n)$ close to the normalized compact operator $\widehat{a}_n(P_0)/\|\widehat{a}_n(P_0)\|$. A scalar bump h supported inside the chart gives the center $\Phi_n^{-1}(4hT)$. The same convexity estimate as in the continuous-trace proof gives a fixed radius $7/2$, while all centers have norm 4.

Proof

We use the general primitive-space obstruction proved in the companion packet [4].

Lemma 1 (primitive π -base obstruction). *If a C^* -algebra B has BCP, then $\text{Prim}(B)$ has a countable π -basis.*

Lemma 2 (countable bump families inside the charts). *For each chart O_n , there is a countable family $\mathcal{B}_n \subset C_c(O_n)$ such that $0 \leq h \leq 1$, $\|h\|_\infty = 1$, and for every nonempty open $W \subset O_n$, some $h \in \mathcal{B}_n$ satisfies $\text{supp } h \subset W$.*

Proof. Since the sets O_m form a π -basis of $\text{Prim}(A)$, the subfamily

$$\{O_m : O_m \subset O_n\}$$

is a π -basis for O_n . The chart identifies O_n with the primitive space of the continuous-trace algebra I_{O_n} , hence O_n is locally compact Hausdorff. For each m with $O_m \subset O_n$, choose

$$h_{n,m} \in C_c(O_n), \quad 0 \leq h_{n,m} \leq 1, \quad \|h_{n,m}\|_\infty = 1, \quad \text{supp } h_{n,m} \subset O_m.$$

Local compactness and regularity give such a bump. The countable family of all $h_{n,m}$'s has the required cofinal support property. $\square$

Proof of Theorem 1. For each n , choose a countable $1/64$ -net

$$D_n \subset S_{\mathcal{K}(H_n)}$$

in the unit sphere of $\mathcal{K}(H_n)$. This is possible because H_n is separable. For $h \in \mathcal{B}_n$ and $T \in D_n$, define

$$c(n, h, T) = \Phi_n^{-1}(4hT) \in I_{O_n} \subset A.$$

The family of all such centers is countable, and

$$\|c(n, h, T)\| = 4.$$

We claim that the balls $B(c(n, h, T), 7/2)$ cover S_A . Since all centers have norm 4, these balls avoid $B(0, 1/2)$, and the claim proves UBCP.

Let $a \in S_A$. The function $P \mapsto \|a + P\|$ on $\text{Prim}(A)$ is lower semicontinuous, so

$$W_a = \{P \in \text{Prim}(A) : \|a + P\| > 3/4\}$$

is nonempty open. Choose n with $O_n \subset W_a$, and choose $P_0 \in O_n$. Put

$$\alpha = \|\hat{a}_n(P_0)\| > 3/4, \quad u = \hat{a}_n(P_0)/\alpha \in S_{\mathcal{K}(H_n)}.$$

Choose $T \in D_n$ with $\|T - u\| < 1/64$.

For $0 \leq s \leq 1$, the map

$$s \mapsto \|\alpha u - 4sT\|$$

is convex. At $s = 0$ the value is $\alpha \leq 1$, while at $s = 1$,

$$\|\alpha u - 4T\| \leq \alpha\|u - T\| + (4 - \alpha)\|T\| < 4 - \alpha(1 - 1/64) < 7/2.$$

Hence

$$\sup_{0 \leq s \leq 1} \|\hat{a}_n(P_0) - 4sT\| < 7/2.$$

By norm-continuity of $\hat{a}_n$, there is a nonempty open neighborhood $N \subset O_n$ of P_0 such that

$$\sup_{0 \leq s \leq 1} \|\hat{a}_n(P) - 4sT\| < 7/2 \quad (P \in N).$$

Choose $h \in \mathcal{B}_n$ with $\text{supp } h \subset N$, and set

$$c = c(n, h, T).$$

For $P \in \text{supp } h$, using $s = h(P)$ gives

$$\|a + P - (c + P)\| = \|\hat{a}_n(P) - 4h(P)T\| < 7/2.$$

For $P \notin \text{supp } h$, the image of c in the primitive quotient at P is zero: if $P \notin O_n$, this is because $c \in I_{O_n} \subset P$; if $P \in O_n \setminus \text{supp } h$, it follows from the displayed section $4hT$. Thus

$$\|a + P - (c + P)\| = \|a + P\| \leq 1 < 7/2.$$

Taking the supremum over primitive quotients gives $\|a - c\| < 7/2$. Therefore a is covered by one of the countably many fixed-radius balls.

This proves UBCP. The implication $\text{UBCP} \Rightarrow \text{BCP}$ is immediate, and Lemma 1 gives $\text{BCP} \Rightarrow$ countable π -basis for every C^* -algebra. Under the chart hypothesis, the three conditions are therefore equivalent. $\square$

Why This Breaks the Previous Wall

The previous failed route tried to use continuous-trace ideals inside a larger type-I algebra. The failure mode was precise: an ambient element may act on such an ideal only as a multiplier, for example $B(H)$ acting on $\mathcal{K}(H)$. The present criterion adds exactly the missing hypothesis, namely that on the selected patches the ambient element is represented by norm-continuous compact operators. Under that hypothesis, the bump-center proof goes through unchanged and does not need a global central $C_0(X)$ -algebra structure.

Scope and Limitations

This is a partial criterion, not a full arbitrary C^* -classification. It applies to any class where one can verify the compact-norming chart π -basis condition. It recovers the continuous-trace/separable-fibre packet by taking local trivial continuous-trace charts. It also explains why unitizations or multiplier algebras with infinite-dimensional fibres remain hard: scalar identity directions are not compact local sections, so they do not satisfy Definition 1 without extra arguments.

Novelty and Search Bounds

The run indexes were searched for arXiv:2311.14257, extension permanence, type-I C^* -algebras, continuous trace, local compact charts, BCP, UBCP, and π -bases. Web searches on June 17, 2026 for BCP permanence, Fell/type-I local continuous-trace structure, and local compact-operator charts found no exact version of this criterion. The proof uses the source problem [1], the companion primitive π -base obstruction [4], and standard continuous-trace local triviality [2, 3].

Human Review Focus

The main point to check is Definition 1: it is intentionally stronger than merely having continuous-trace ideals. Once the chart representatives $\hat{a}_n$ are available for all $a \in A$, the proof is the same fixed-constant bump argument as in the continuous-trace packet. Reviewers should also check the use of lower semicontinuity of $P \mapsto \|a + P\|$ and the fact that $c \in I_{O_n}$ vanishes in primitive quotients outside O_n .

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] Jacques Dixmier, *C^* -algebras*, North-Holland, 1977.
- [3] Iain Raeburn and Dana P. Williams, *Morita Equivalence and Continuous-Trace C^* -Algebras*, American Mathematical Society, 1998.
- [4] Run `fa.banach_001`, lane 19, *A Primitive-Space π -Base Obstruction for C^* -Algebras with BCP*, solution packet `2311.14257_primitive_pi_base_obstruction`, 2026.
- [5] Run `fa.banach_001`, lane 19, *BCP for Continuous-Trace C^* -Algebras with Separable Fibres*, solution packet `2311.14257_continuous_trace_separable_fiber_bcp`, 2026.